Day 7 — Nārāyaṇa Paṇḍita and *Gaṇita-kaumudī*

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) place Nārāyaṇa Paṇḍita (14th c. CE) in the chronology of Indian mathematics, (b) describe what the *Gaṇita-kaumudī* covers, and (c) write a short reflection on why systematic written method matters.

Materials

- Whiteboard
- Notebook
- A printed simple chronology chart (handout, see Student-facing content)

Vocabulary review

- *Gaṇita-kaumudī* (introduced in `_shared/glossary.md`) — "Moonlight of Mathematics," Nārāyaṇa's 1356 CE treatise.
- *Bhadragaṇita* — Book 14 of the *Gaṇita-kaumudī*, devoted to magic squares.

Lesson flow

Warm-up (5 min)

- *"You've been using Nārāyaṇa Paṇḍita's algorithm for three days. Today: who was he, what did he write, and why does it matter?"*
- Quick free-write (60 seconds): "What do I think I know about Nārāyaṇa already?" — students jot. Not collected.

Core (20 min)

1. **Chronology.** Build a simple timeline on the board:

Period	Indian mathematics
~500 BCE	*Śulba Sūtras* — geometric ritual texts
~500 CE	Āryabhaṭa (b. 476) — *Āryabhaṭīya*
~600 CE	Brahmagupta — *Brāhmasphuṭasiddhānta*
~1100 CE	Bhāskara II — *Līlāvatī*, *Bījagaṇita*
1356 CE	**Nārāyaṇa Paṇḍita — *Gaṇita-kaumudī***
~1400 CE	Mādhava (Kerala) — early infinite series

“Nārāyaṇa wrote in the 14th century — relatively ‘late’ in the medieval Indian mathematical tradition, but he produced one of the most comprehensive single works.”

2. **What is in the *Gaṇita-kaumudī*?** Fourteen books (*vyavahāras*):
    - Book 1–3: arithmetic (basic operations, fractions)
    - Book 4–6: ratios, percentage problems, alligation
    - Book 7–8: progressions and series
    - Book 9: combinatorics (permutations, partitions — Nārāyaṇa was a pioneer here)
    - Book 10–11: geometry of plane figures and solids
    - Book 12: triangular and figurate numbers
    - Book 13: practical problems (e.g. dilution of milk, mixture problems)
    - **Book 14: *Bhadragaṇita* — magic squares**

    “The treatise is comprehensive — it isn’t only about magic squares. But the magic-square book is the longest and most systematic in any pre-modern work we have.”

3. **Why systematic written method matters.** Discuss together:
    - Before Nārāyaṇa, magic squares appeared in many Indian and non-Indian sources — but mostly as *examples*, not as recipes.
    - Nārāyaṇa gives **methods**: rules that work for any input size. This is what makes it mathematics, not just a curiosity collection.
    - This is the same distinction we draw today between “here’s a Sudoku puzzle” and “here’s an algorithm to generate Sudoku puzzles.”
4. **A paraphrase from the source.** From *Bhadragaṇita*, paraphrased:

    “Magic squares of odd order are made by placing the first number in the top middle cell, and proceeding diagonally upward and rightward; if blocked, descend by one.” (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* Book 14, 1356 CE — paraphrased.)

    This is the rule you have been using. The original is in Sanskrit; this is a faithful paraphrase, not a quoted verse.

Activity (15 min) — Reflection paragraph

- Each student writes 5–7 sentences in response to:

    “Nārāyaṇa Paṇḍita gave us a rule, not a magic square. Why is having a rule more valuable than having just one example? Use specific evidence from this module.”

- Pairs swap and read each other’s paragraphs. One quick comment each: “the part that landed” and “what I’d want more of.”

Wrap-up (5 min)

- Two volunteer-read paragraphs (with permission).
- Preview Day 8: *“Tomorrow we go cross-cultural — China’s Lo Shu and Europe’s Dürer.”*

Student-facing content

Nārāyaṇa Paṇḍita (14th c. CE) was a mathematician of medieval India who wrote the *Gaṇita-kaumudī* (“Moonlight of Mathematics”) in **1356 CE**. The work has 14 books (*vyavahāras*) covering arithmetic, algebra, geometry, combinatorics, and recreational mathematics.

The 14th and final book is *Bhadragaṇita* (“Mathematics of Auspicious Squares”). It is the most comprehensive pre-modern treatment of magic squares in any tradition, giving:

- The magic-constant formula $S = n(n^2 + 1)/2$ (which you derived on Day 2).
- The stair-step (*turagagati*) method for any odd-order square (which you used on Days 4–5).
- Complement-based methods for even-order squares.
- Methods for *singly-even* orders (6, 10, 14, …).
- Discussion of equivalence and symmetry classes.

Paraphrase from the source: “Magic squares of odd order are made by placing the first number in the top middle cell, and proceeding diagonally upward and rightward; if blocked, descend by one.” (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* Book 14, 1356 CE — paraphrased.)

Having a *rule* is more powerful than having an *example*. A rule generates infinitely many squares. An example gives you one.

Homework

- Polish your reflection paragraph from class. Add one more specific example (e.g. “I used the rule today to construct a 7×7 square in under 5 minutes; without the rule I would have been guessing all night.”).

Differentiation

- **One tier up:** Compare Nārāyaṇa to Bhāskara II (~1150 CE). What did Bhāskara cover that Nārāyaṇa didn’t? Vice versa? (Bhāskara: calculus precursors, planetary motion. Nārāyaṇa: combinatorics, magic squares.)
- **One tier down:** Write just 3–4 sentences. Focus on the “rule vs. example” distinction.

Teacher notes

- Avoid hero-worship framing. Nārāyaṇa was a brilliant systematiser; he was not the sole discoverer of magic squares.
- A few students will ask whether Nārāyaṇa’s methods were known outside India. Answer: yes, eventually — the same odd-order method shows up in 17th-c. European mathematics (via de la Loubère, 1687) probably with intermediate Islamic-world transmission. The independent re-discovery story is interesting but speculative.
- The reflection paragraph is a low-stakes formative — don’t grade harshly. Read for understanding, not for polish.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — primary source paraphrased.
- Plofker (2009), *Mathematics in India*, ch. on 14th-c. mathematics — secondary scholarly context.